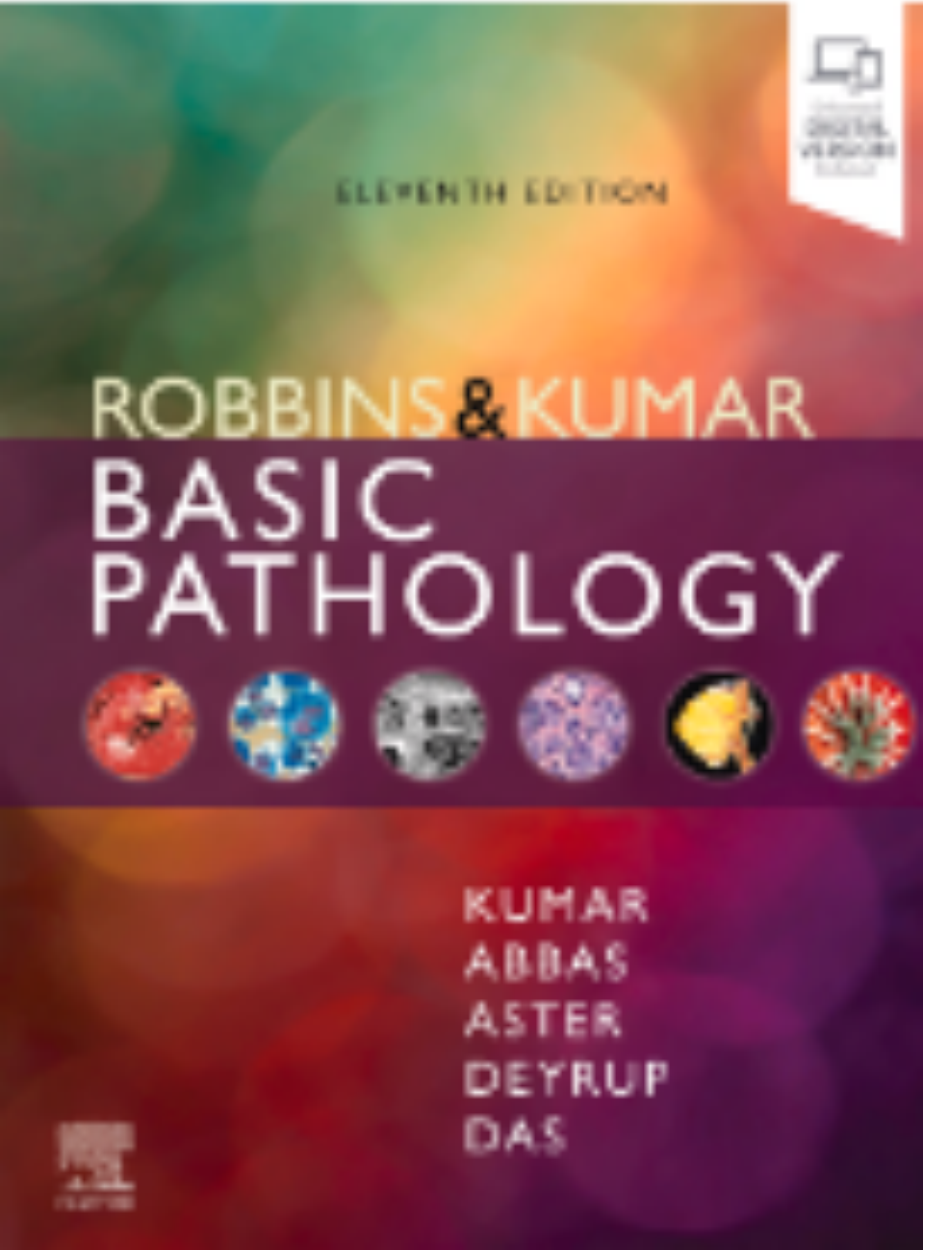


Diseases of Myelin

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ELEVENTH EDITION
ROBBINS & KUMAR
BASIC PATHOLOGY
KUMAR
ABBAS
ASTER
DEYRUP
DAS


NO PHOTOGRAPHY
NO VIDEO RECORDING

Learning Objectives

- Describe leukodystrophies and their key features
- Discuss the most common demyelinating diseases of the peripheral nervous system, including their causes, clinical features, and pathological findings

Myelin Oligodendrocyte Glycoprotein Antibody-associated Disease (MOGAD)

- An inflammatory disorder of the central nervous system characterized by attacks of immune-mediated demyelination predominantly targeting the optic nerves, brain, and spinal cord
- The disease has a predilection for children
- MOG antibodies
  - Myelin oligodendrocyte glycoprotein (MOG), a protein on the surface of oligodendrocytes

وهي الخلايا المسؤولة عن تصنيع المايلين
داخل الجهاز العصبي المركزي.

1. Myelin Oligodendrocyte Glycoprotein Antibody-associated Disease (MOGAD)

- MOGAD may have a monophasic or relapsing course
- Attacks usually develop over days and may plateau with variable recovery over weeks to months
- Attacks may be preceded by an infectious illness or vaccination
- Clinical features:
  - Optic neuritis, acute disseminated encephalomyelitis (ADEM), transverse myelitis, or other central nervous system (CNS) manifestations, either alone or in combination

2. Acute disseminated encephalomyelitis (ADEM)

- Known as postinfectious encephalomyelitis
- Clinical syndrome characterized by a first polyfocal CNS episode from presumed demyelination that includes encephalopathy not explained by fever, systemic illness, or postictal features
- Acute multifocal neurologic deficits and often rapidly progressive (nonlocalizing, rapidly progressive symptoms, headache, lethargy, coma), unlike MS
  - Fatal in ~20% of cases; most survivors recover fully
- Can be MOG antibodies positive (MOGAD) or negative

3. Neuromyelitis Optica Spectrum Disorder (NMOSD)

- An antibody-mediated demyelinating disease centered on the optic nerves and spinal cord
  - Other areas can be involved
- Positive for serum NMO-immunoglobulin G (IgG) antibody that selectively binds aquaporin-4 (AQP4)
  - (AQP4) receptors, predominate in astrocytes rather than oligodendrocytes
    - Unlike multiple sclerosis, necrosis and cavitation typically involve both gray and white matter



4. Osmotic Demyelination Syndrome

- Caused by nonimmune damage to oligodendrocytes
- Typically after sudden correction of hyponatremia
- Rapid demyelination in the central pons (most cases) and sometimes other brain regions due to rapid shifts in serum osmolality
- Rapidly evolving quadriplegia

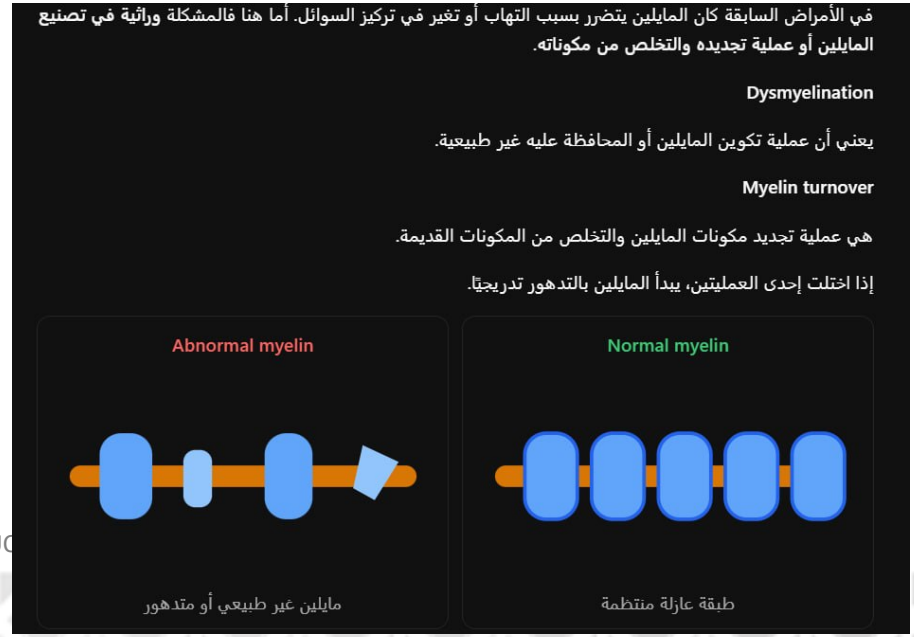
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5. Leukodystrophies

مشكلة وراثية بتصنيع المايلين

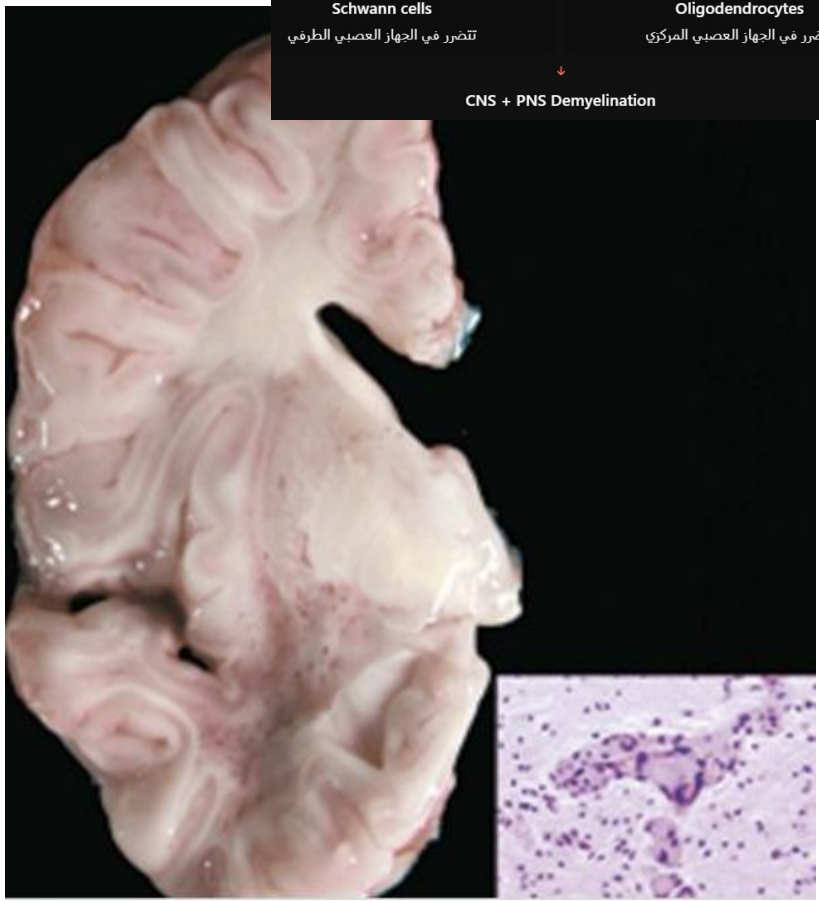
- Inherited dysmyelinating diseases characterized by abnormal myelin synthesis or turnover
- In contrast to MS, the neurologic defects present at an early age and are progressive
- Imaging studies reveal diffuse and symmetric myelin loss.



6. Krabbe Disease

- AR Lysosomal storage disorder
- Resulting from a deficiency of specific enzyme →accumulation of galactolipids→ toxic to oligodendrocytes and schwann cells→ demyelination in the brain and peripheral nerves and loss of oligodendrocytes in the CNS
- Survival beyond 2 years is rare





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Globoid cells

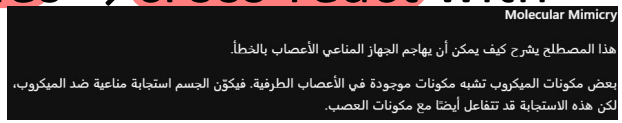
7. Metachromatic Leukodystrophy

- AR an autosomal recessive disease
- Deficiency of the lysosomal enzyme
- The prognosis depends on the age at diagnosis, with earlier age of onset associated with more rapid progression.
- Hematopoietic stem cell transplantation, which may allow repopulation of the CNS with enzymatically competent macrophages, has shown some benefit in Krabbe disease and metachromatic leukodystrophy.

7 6 7

8. Guillain-Barré Syndrome (GBS)

- A rapidly progressive acute demyelinating disorder affecting motor axons, resulting in ascending weakness يبدا من تحت القدمين الى فوق اليدين
- Symptoms worsen over ~2 weeks, most patients reaching peak severity by 4 weeks.
- One of the most common life-threatening diseases of the peripheral nervous system ✓✓✓
- 2/3rd of cases are triggered by an infection→ generation of microbe-specific T cells and antibodies →cross-react with antigens in the nerve sheath.



  - Campylobacter jejuni, Epstein-Barr virus, cytomegalovirus, (HIV), Zika virus, and SARS-CoV-2.

Guillain-Barré Syndrome

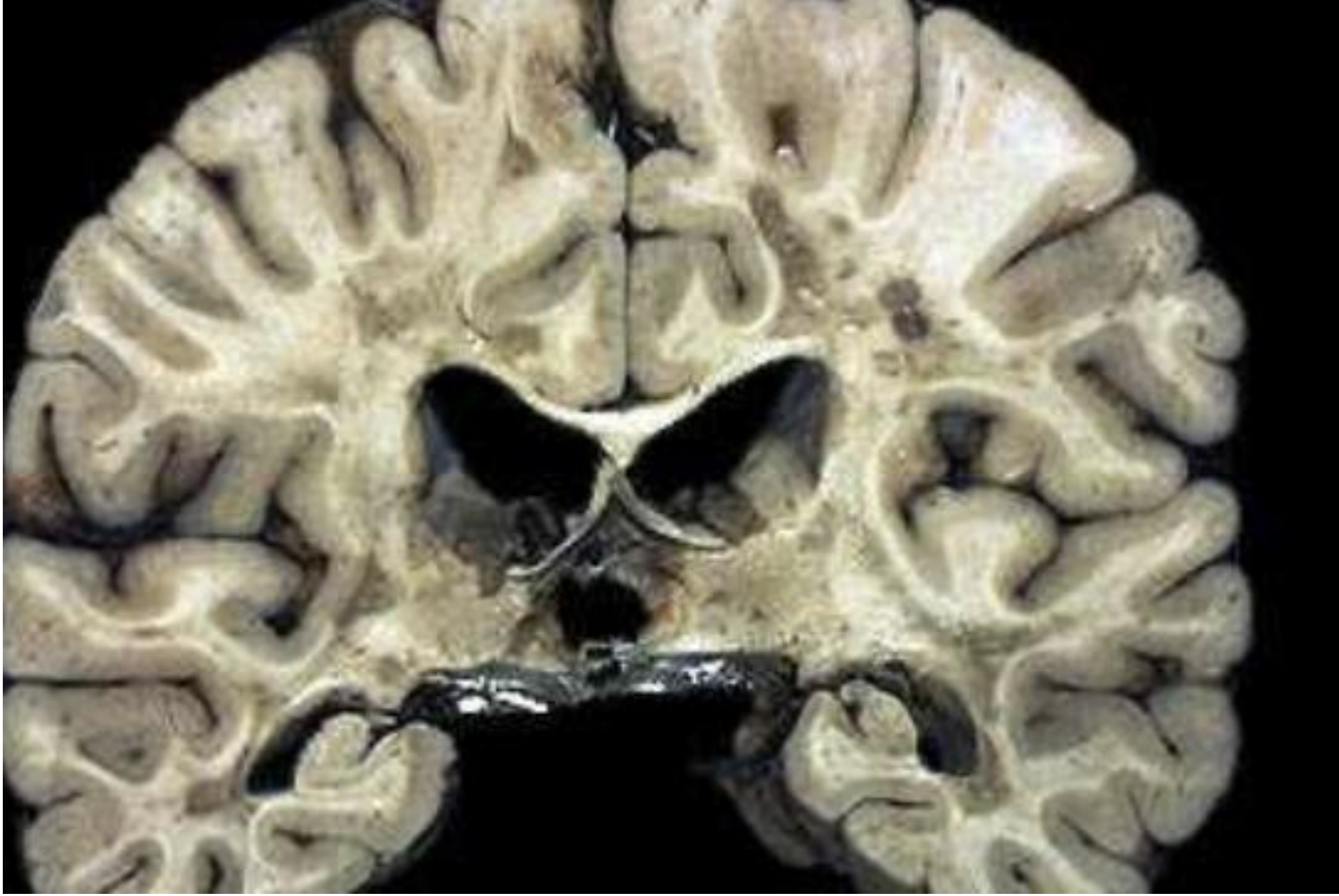
- The injury is most extensive in the nerve roots and proximal nerve segments
- Associated with mononuclear cell infiltrates rich in macrophages
- Treatments include:
  - Plasmapheresis (to remove offending antibodies)
  - Intravenous immunoglobulin infusions
  - Supportive care, such as ventilatory support
  - Patients who survive the initial acute phase of the disease usually recover with time

9 Chronic Inflammatory Demyelinating Polyneuropathy (CIDP)

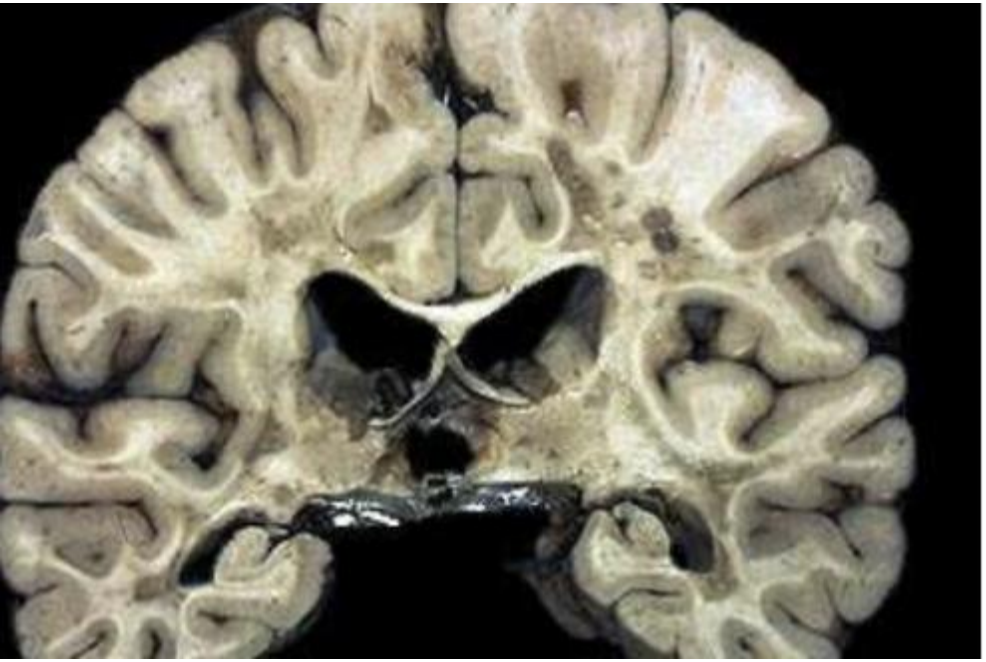
- An inflammatory peripheral neuropathy, Characterized by symmetrical mixed sensorimotor polyneuropathy that progresses for 2 months or more
- Both motor and sensory abnormalities are common
  - Weakness, difficulty in walking, numbness, pain, and tingling
- Like GBS, CIDP is immune mediated
- Unlike GBS, chronic, relapsing-remitting, or progressive course
- ↑ in patients with paraproteinemias, lymphoid neoplasms, and HIV infection
- Nerves show segments of demyelination and remyelinatio

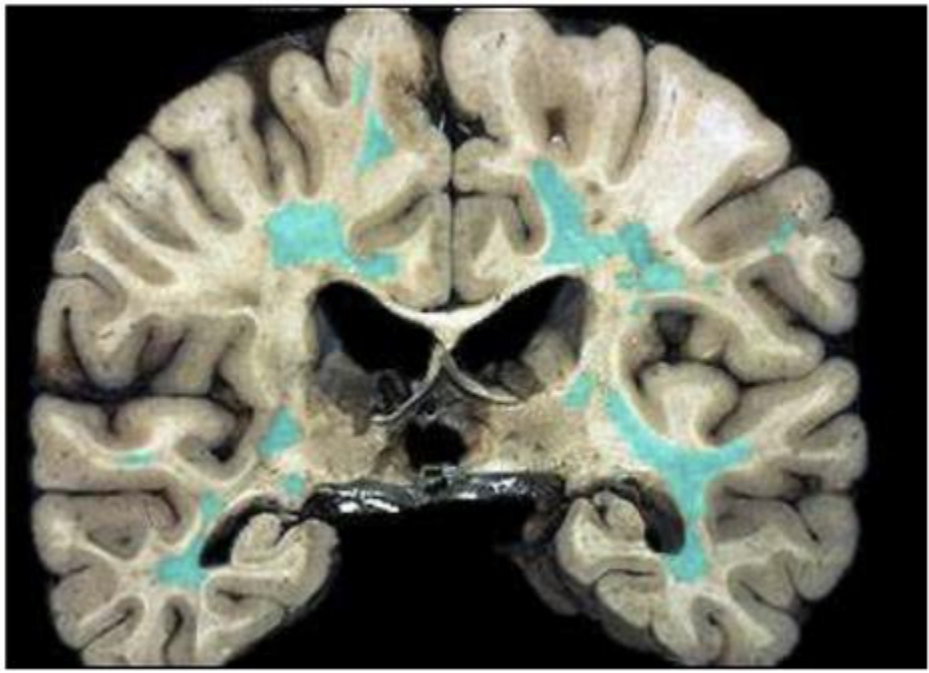


Self study



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